DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW



Evaluation Scheme & Syllabus for
B. Tech. 3rd Year

Automobile Engineering

ON CBCS

[Effective from session 2018-19]

DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

AUTOMOBILE ENGINEERING

EVALUATION SCHEME (CBCS)

V SEMESTER

Sl No.	Subject Code	Subject Name	Teaching Deptt.	L-T-P	Th/Lab Marks	Sessional		Total	Credit
						Test	Assig/ Att		
1	RAS501	MANEGIRIAL ECONOMICS	Applied Science	3---0---0	70	20	10	100	3
2	RAS 502/ RUC501	SOCIOLOGY /CYBER SECURITY	Applied Science	3---0---0	70	20	10	100	3
3	RME504	Design of Machine Elements	ME.	3---0---0	70	20	10	100	3
4	RME502	Heat & Mass Transfer	ME.	3---1---0	70	20	10	100	4
5	RAU502	Automotive Engines	ME	3---0---0	70	20	10	100	3
6	RME055/ RAU052/ RME052/ RME-053	**Department Elective Course –1**	ME	3---1---0	70	20	10	100	4
7	RME554	Design & Simulation Lab	ME	0---0---2	50		50	100	1
8	RME552	Heat & Mass Transfer Lab	ME	0---0---2	50		50	100	1
9	RAU552	Automotive Components Lab	ME	0---0---2	50		50	100	1
10	RAU559	Seminar – I		0--0--2	50		50	100	1
	TOTAL							**1000**	**24**

Department Elective Course –1

RME055: Advanced Manufacturing Science
RAU052: Vehicle Transport Management
RME052: Mechatronics & Microcontroller
RME053: Finite Element Analysis

DESIGN OF MACHINE ELEMENTS

UNIT I

Introduction

Design requirements of machine elements, Principles of mechanical design, Modes of failures and Factor of Safety. Systematic design process, Aesthetic and Ergonomic considerations in design, Use of standards in design, Manufacturing consideration in design, Selection of preferred sizes, Indian Standards designation, Designation of carbon & alloy steels, Selection of materials for static and fatigue loads.

UNIT II

Design for Static and Fluctuating Loads

Cyclic stresses, Fatigue and endurance limit, Notch sensitivity, Stress concentration. Design for finite and infinite life, Soderberg, Goodman & Gerber criteria.

Shafts, Keys and Couplings

Materials for shaft, Stresses in shafts, Design of shafts subjected to twisting moment, bending moment and combined twisting and bending moments, Shafts subjected to fatigue loads.
Types of keys, splines, Selection of keys, Strength of keys, Design of rigid and flexible couplings.

UNIT III

Design of Mechanical Drives

Introduction to power transmission and drives. Classification of gears, Terminology, Gear tooth proportions, Tooth forms, System of gear teeth, contact ratio, Standard proportions of gear systems, Interference in involute gears, Backlash, Selection of gear materials, Gear manufacturing methods, Failure of gear tooth, Design considerations, AGMA and Indian standards, Beam strength and wear strength of gear tooth, Design of spur and helical gears.

UNIT IV

Mechanical Springs

Classification of springs, Material for helical springs, End connections for compression and tension helical springs, Design of helical springs subjected to static and fatigue loading.

Power Screws

Forms of threads, multiple threads, Efficiency of square threads, Trapezoidal threads, Stresses in screws, Design of screw jack.

UNIT V

Basic of Finite Element Method, Variational calculus, Integral formulation, variational methods: Methods of weighted residuals, Approximate solution using variational method, Modified Galerkin method, Boundary conditions.

Basic Finite Element Concepts: Basic ideas in a finite element solution, General finite element solution procedure, Finite element equations using modified Galerkin method, Axis symmetric Problems, Axial bars, Torsion bars, Application in Solid Mechanic Problems, Plane truss problems using software application.

Note: Design data book is allowed in the examination

Text Books:

1. Design of Machine Elements,V.B. Bhandari, Tata McGraw Hill Co.
2. Design of Machine Elements, Sharma and Purohit, PHI.
3. Machine Design, P C Gope, PHI Publication
4. Machine Design, Sadhu Singh, Khanna Publishing House
5. Machine Design-R S Khurmi and J K Gupta, S Chand
6. J N Reddy "An Introduction to finite element method" Tata Mc Graw Hill 3rd edition
7. Machine Design Data Book, Sadhu Singh, Khanna Publishing House
8. S.S. Rao, "Finite Element Method In Engineering", Pergaman Press
9. P Seshu, Finite Element Analysis, PHI publications, Delhi
10. Design data book: Design Data Handbook for Mechanical Engineering in SI and Metric Units – by K. Mahadevan, and K. Balaveera Reddy.

References:

1. Design of Machine Elements-M.F. Spott, Pearson Eductaion
2. Machine Design-Maleev and Hartman, CBS Publishers.
3. Mechanical Engineering Design, 9e – Joseph E. Shigely, McGraw Hill Education.
4. Design of Machine Elements, Gope PHI.
5. Finite Element Method with Applications in Engineering Y M Desai, Pearson Publication
6. Introduction to Finite Element Analysis by Tirupathi R.Chandrupatla & Ashok D Belegundu, Pearson Publication
7. V.Ramamurti "Finite Element Method in Machine Design"Norosa Publishing House

HEAT & MASS TRANSFER

UNIT-I

Introduction to Heat Transfer: Thermodynamics and Heat Transfer. Modes of Heat Transfer: Conduction, convection and radiation. Effect of temperature on thermal conductivity of materials; Introduction to combined heat transfer mechanism. 2 Conduction : General differential heat conduction equation in the rectangular, cylindrical and spherical coordinate systems.Initial and boundary conditions. 3 Steady State one-dimensional Heat conduction : Simple and Composite Systems in rectangular, cylindrical and spherical coordinates with and without energy generation; Concept of thermal resistance.Analogy between heat and electricity flow; Thermal contact resistance and over all heat transfer coefficient; Critical radius of insulation.

UNIT-II

Fins: Heat transfer from extended surfaces, Fins of uniform cross-sectional area; Errors of measurement of temperature in thermometer wells. 3 Transient Conduction: Transient heat conduction; Lumped capacitance method; Time constant; Unsteady state heat conduction in one dimension only, Heisler charts.

UNIT-III

Forced Convection: Basic concepts; Hydrodynamic boundary layer; Thermal boundary layer; Approximate integral boundary layer analysis; Analogy between momentum and heat transfer in turbulent flow over a flat surface; Mixed boundary layer; Flow over a flat plate; Flow across a single cylinder and a sphere; Flow inside ducts; Thermal entrance region, Empirical heat transfer relations; Relation between fluid friction and heat transfer; Liquid metal heat transfer. Natural Convection : Physical mechanism of natural convection; Buoyant force; Empirical heat transfer relations for natural convection over vertical planes and cylinders, horizontal plates and cylinders, and sphere, Combined free and forced convection.

UNIT-IV

Thermal Radiation : Basic radiation concepts; Radiation properties of surfaces; Black body radiation Planck’s law, Wein’s displacement law, Stefan Boltzmann law, Kirchoff’s law; ; Gray body; Shape factor; Black-body radiation; Radiation exchange between diffuse non black bodies in an enclosure; Radiation shields; Radiation combined with conduction and convection; Absorption and emission in gaseous medium; Solar radiation; Green house effect.

UNIT-V

Heat Exchanger : Types of heat exchangers; Fouling factors; Overall heat transfer coefficient; Logarithmic mean temperature difference (LMTD) method; Effectiveness-NTU method; Compact heat exchangers. 3 Condensation and Boiling: Introduction to condensation phenomena; Heat transfer relations for laminar film condensation on vertical surfaces and on outside & inside of a horizontal tube; Effect of non-condensable gases; Dropwise condensation; Heat pipes; Boiling modes, pool boiling; Hysteresis in boiling curve; Forced convection boiling. 3 Introduction to Mass Transfer: Introduction; Fick's law of diffusion; Steady state equimolar counter diffusion; Steady state diffusion though a stagnant gas film.

Books:

1. Fundamentals of Heat and Mass Transfer, by Incroperra& DeWitt, John Wiley and Sons
2. Heat and Mass Transfer by Cengel, McGraw-Hill
3. Heat Transfer by J.P. Holman, McGraw-Hill
4. Heat and Mass Transfer by Rudramoorthy and Mayilsamy, Pearson Education
5. Heat Transfer by Ghoshdastidar, Oxford University Press
6. A text book on Heat Transfer, by Sukhatme, University Press.
7. Heat Transfer by Venkateshan, Ane Books Pvt Ltd
8. Schaum's outline of Heat Transfer by Pitts & Sisson McGraw-Hill
9. Heat and Mass Transfer by R Yadav, Central Publishing House

AUTOMOTIVE ENGINES

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Unit-I

Construction and operation: Constructional details of spark ignition (SI) and compression ignition (CI) engines. Working principles.Two stroke SI and CI engines – construction and working. Comparison of SI and CI engines and four stroke and two stroke engines. Engine classification, firing order. Otto, diesel and dual cycles.

Unit–II

SI Engines: Combustion in SI engine, Flame speed, Ignition delay, abnormal combustion and it's Control, combustion chamber design for SI engines,Carburetion, Mixture requirements, Carburetors and fuel injection system in SI Engine, Ignition system requirements, Magneto and battery ignition systems, ignition timing and spark Plug, Electronic ignition, Scavenging in 2 Stroke engines, Supercharging and its effect

Unit–III

CI Engine: Combustion in CI engines, Ignition delay, Knock and it's control, Combustion chamber design of CI engines.

Fuel injection in CI engines, Requirements, Types of injection systems, Fuel pumps, Fuel injectors, Injection timings ,Exhaust emissions from SI engine and CI engine and it's control

Unit-IV

Engine Cooling and Lubrication: Different cooling systems, Radiators and cooling fans, Engine Friction, Lubrication principle, Type of lubrication, Lubrication oils, Crankcase ventilation. Different Fuels used in SI & CI Engines.

Testing and Performance: Performance parameters, Basic measurements, Blow by measurement, Testing of SI and CI engines

Unit V

Compressors: Classification, Reciprocating compressors, Single and Multi stage compressors, Intercooling, Volumetric efficiency.

Rotary compressors, Classification, Centrifugal compressor , Axial compressors, Surging and stalling, Roots blower, Vaned compressor.

TEXT BOOKS:

1.Ganesan V., "Internal Combustion Engines", Tata McGraw Hill, 2007

2 Ramalingam K.K., “Internal Combustion Engines”, Sci-Tech Publications, 2005

3. Mathur and Sharma "Fundamental Combustion Engines" Dhanpat Rai and Sons, 2002 Engineering

4. A.K. Babu, Automotive Engines, Khanna Publishing House

4.Fundamentals of Internal Combustion Engines by W.W. Pulkrabek,,Pearson Education

5. I.C Engine, by R. Yadav, Central Publishing House, Allahabad

REFERENCES:

1. Heisler, "Advanced Engine Technology" SAE Publication, 1995

2. Edward F. Obert "Internal Combustion Engines" 3 Edition, 1970

3. Gupta. H.N. "Fundamentals of Internal Combustion" Engines, reprint, PHI Learning Pvt. Ltd. 2006

4. Fundamentals of Internal Combustion Engine by Gill, Smith,Ziurs, Oxford & IBH Publishing CO

5. AK Babu, Automotive Electrical and Electronics, Khanna Publishing house.
6. John B. Heywood, “Fundamentals of Internal Combustion Engines”, 1988

DESIGN AND SIMULATION LAB

1. Design & Modeling of Cotter joint.
2. Design & Modeling of Knuckle joint
3. Design & Modeling of riveted joint applied to boiler joints.
4. Study of a FEA package and modeling stress analysis of
   a. Bars of constant cross section area, tapered cross section area and stepped bar
   b. Trusses,
   c. Beams – Simply supported, cantilever, beams with UDL, beams with varying load etc

5. Mini Project: Will take up problems from real life applications and optimize using modeling and analysis software

   Each student will be given a real life problem for the complete design of a subsystem/system using either manual calculation with the help of design handbook or through computer programme or through software tools. This assignment should be done in groups, which will be submitted at the end of the semester.

Text Books:

1. Design of Machine Elements, V.B. Bhandari, Tata McGraw Hill Co.
2. Design of Machine Elements, Sharma and Purohit, PHI.
3. Machine Design-R S Khurmi and J K Gupta, S Chand
4. Machine Design, Sadhu Singh, Khanna Publishing House
5. J N Reddy "An Introduction to finite element method" Tata Mc Graw Hill 3rd edition
6. S.S. Rao, "Finite Element Method In Engineering", Pergaman Press
7. Machine Design, Sadhu Singh, Khanna Publishing House
8. P Seshu, Finite Element Analysis, PHI publications, Delhi

Design data book: Design Data Handbook for Mechanical Engineering in SI and Metric Units – by K. Mahadevan, and K. Balaveera Reddy

References:

1. Design of Machine Elements-M.F. Spott, Pearson Eductaion
2. Machine Design-Maleev and Hartman, CBS Publishers.
3. Mechanical Engineering Design, 9e – Joseph E. Shigely, McGraw Hill Education.
4. Design of Machine Elements, Gope PHI.
5. Finite Element Method with Applications in Engineering Y M Desai, Pearson Publication
6. Introduction to Finite Element Analysis by Tirupathi R.Chandrupatla & Ashok D Belegundu, Pearson Publication
7. V.Ramamurti "Finite Element Method in Machine Design"Norosa Publishing House

HEAT & MASS TRANSFER –LAB

Minimum eight experiment of the following

1. Conduction – Experiment on Composite plane wall
2. Conduction – Experiment on Composite cylinder wall
3 Conduction - Experiment on critical insulation thickness
4. Conduction – Experiment on Thermal Contact Resistance
5. Convection - Pool Boiling experiment
6. Convection - Experiment on heat transfer from tube-(natural convection).
7. Convection - Heat Pipe experiment.
8. Convection - Heat transfer through fin-(natural convection) .
9. Convection - Heat transfer through tube/fin-(forced convection).
10 Convection - Determination of thermal conductivity of fluid
11 Experiment on Stefan's Law, on radiation determination of emissivity, etc.
12 Experiment on solar collector, etc.
13. Heat exchanger - Parallel flow experiment
14. Heat exchanger - Counter flow experiment

AUTOMOTIVE COMPONENTS LAB

LIST OF EXPERIMENTS (Minimum Ten Experiments are required to be conducted)

1. Dismantling and study of Multi-cylinder Petrol Engine
2. Assembling of Multi-cylinder Petrol Engine
3. Dismantling and study of Multi-cylinder Diesel Engine
4. Assembling of Multi-cylinder Diesel Engine
5. Study of petrol engine fuel system
6. Study of diesel engine fuel system
7. Study and measurement of light and heavy commercial
8. Vehicle Frame
9. Study, dismantling and assembling of front and rear
10. Axles
11. Study, dismantling and assembling of differential
12. Study, dismantling and assembling of Clutch
13. Study, dismantling and assembling of Gear Box
14. Study of steering system
15. Performance of CI and SI engine
16. Impact of Variable compression ratio on Performance

ADVANCED MANUFACTURING SCIENCE

UNIT I

Introduction: Introduction to Advanced Manufacturing processes and its classification.

Mechanical Type Material Removal Processes: Ultrasonic machining; Elements of the process; Tool design and economic considerations; Applications and limitations, Abrasive jet and Abrasive water jet machining principles; Mechanics of metal removal; Design of nozzles; applications, Abrasive finishing process, Magnetic abrasive finishing process.

UNIT II

Thermal Type Material Machining Processes: Classification, General principles and applications of Electro discharge, Plasma arc, Ion beam, Laser beam, Electron beam machining, Mechanics of metal removal in EDM, selection of EDM pulse generator dielectric, machining accuracy, surface finish and surface damage in EDM, Generation and control of electron beam for machining applications, advantages and limitations

UNIT III

Chemical and Electro-chemical Type Material Removal Processes: Principle, working advantages, disadvantages and applications of Electrochemical, Chemical machining, Economy aspects of ECM, Electro-chemical deburring and honing.

UNIT IV

Hybrid Machining Processes: Introduction and working Principal of ECDM, ECAM, Abrasive EDM, Micro EDM etc.

UNIT V

Unconventional Forming processes: Principle, working and applications of High energy forming processes such as Explosive Forming, Electromagnetic forming, Electro Discharge forming, water hammer forming, explosive compaction etc.

Electronic-device Manufacturing: Brief description of Diffusion and Photo- Lithography process for electronic-device manufacturing

Books:

1. Advance Machining Processes V.K. Jain New Age
2. Modern Machining Processes P.C. Pandey New Age
3. Manufacturing Processes Degarmo -
4. Manufacturing Processes Kalpakjian Tata McGraw-Hill International

VEHICLE TRANSPORT MANAGEMENT

UNIT-I

Historical Back ground: Introduction, the growth of a network, trams, trolley buses, private car's subsidies The Infrastructure: Road- Approach Road. Highways National, State, District, traffic condition, relief of congestion, pedestrians, zebra lines, margins, shopping centres. Bus-stops, shelters.Bus stations.Garages layout of premises, equipment, use of machinery, conveyance of staff, facilities for passengers. Maintenance -preventive, breakdown, overhauling -major, minor.

UNIT –II

Organisation and Management: Forms of ownership, principle of transport, management – internal organisation, centralised condition, decentralised condition (Engineering, traffic and administration), staff administration: industrial relation, administration, recruitment and training, welfare, health and safety.

Public relations divisions: Dissemination of information, maintaining goodwill- handling complaints, traffic advisory, committees- local contractors co-operation with the press news and articles- facilities for visitors- forms of publicity importance of quality -inter departmental liaison advertisements, signs, notice and directions general appearance of premises, specialized publicity.

UNIT-III

Prevention of accidents: Emphasis of safe driving-annual awards bonus encouragement vehicle design platform, layout, location of steps, scheduled route hazards records elimination of accident prone devices.

Route planning: Source of traffic, town planning, turning paints, stopping places, shelters survey of route preliminary schedule test runs elimination of hazards factors affecting. Frequency direction of traffic flow estimated traffic possibility single verses double deck.

UNIT-IV

Timing, bus working and schedules: Time table layout uses of flat graph method of presentation preparation of vehicle and crew schedule preparation of the duty roster, co-operation with employers use of the vehicle running numbering determination of vehicle efficiency, checking efficiency of crew, duty arrangements. Fare collections systems: Principles of collection the way bill, bell punch system reduced ticket stocks wilk brew system T.I.M and straight M/C/S. The verometerlensonparason coach tickets exchanges, box system personal and common stock flat fare platform control.

UNIT-V

The fare structure: Basis of fares historical background effects of competition and control calculating average zone system straight and tapered scale elastic and inelastic demand coordination of fares concessions fares changes for workman. Anomalies double booking inter availability through booking and summation private hire charges. Operating cost and types of vehicles: Classification costs, average speed running costs supplementary costs depreciation obsolescence, life of vehicles sinking fund factor affecting post per vehicles mile incidence of wages and overheads 100 seats miles basis, average seating capacity vehicles size and spread overs, types of vehicle economic considerations authorization of trolley, bus services, statuary procedure taxes and hire cars.

TEXTBOOKS:

1. Bus operation -L.d kitchen, iliffe& sons
2. Bus & coach operation -Rex w. fautks. butterworth version of 1987
3. Hybrid Electric Vehicles, AK Babu, Khanna Publishing House

FINITE ELEMENT METHOD

UNIT-I

Introduction Introduction to finite difference method and finite elements method, Advantages and limitations, Mathematical formulation of FEM, Different approaches in Finite Element Method - Direct Stiffness approach, simple examples, Variational approach, Elements of variational calculus - Euler Lagrange equation, Rayliegh Ritz method, Weighted Residual methods, Point Collocation method, Galarkin method - Steps involved in FEM.

UNIT-II

Types of Elements Used Interpolation Polynomials - Linear elements Shape function - Analysis of simply supported beam - Element and Global matrices - Two-dimensional elements, triangular and rectangular elements - Local and Natural Co-ordinate systems.

UNIT-III

Finite Element Formulation of Field Problems 1-D and 2-D heat transfer, fluid flow (incompressible and non viscous fluid) in ducts, Simple electrical and magnetic field problems. Simple Numerical examples

UNIT-IV

Finite Element Formulation of Solid Mechanics Problems 1-D problem of shaft; Truss element analysis of pinned truss, Plane stress/strain problems, Axi-symmetric problems, thin plate problems; Vibration of shafts & beams.

UNIT-V

Numerical Methods in FEM Evaluation of shape functions - One dimensional & triangular elements, Quadrilateral elements, Isoperimetric elements - Numerical Integration, Gauss Legendre quadrature - Solution of finite element equations – Gauss Elimination Method, Cholesky decomposition.

Books:

1. The Finite Element Method O.C. Zienkiewicz and R.L. Taylor McGraw Hill
2. An Introduction to Finite Element Method J. N. Reddy McGraw Hill
3. Finite Element Procedure in Engineering Analysis K.J. Bathe McGraw Hill
4. Finite Element Analysis C.S. Krishnamoorthy Tata McGraw Hill
5. Concepts and Application of Finite Element Analysis R.D. Cook, D.S. Malcus and M.E. Plesha John Wiley
6. Introduction to Finite Elements in Engineering T.R Chandragupta and A.D. Belegundu Prentice Hall India
7. Finite Element and Approximation O.C. Zenkiewicy & Morgan –
8. Numerical Methods E Balagurusamy Tata McGraw Hill

MECHATRONICS AND MICROPROCESSOR

UNIT - I

Introduction to Mechatronic Systems: Measurement and control systems their elements and functions, Microprocessor based controllers.

Review of Transducers and Sensors: Definition and classification of transducers. Definition and classification of sensors. Principle of working and applications of light sensors, proximity sensors and Hall effect sensors.

UNIT - II

Electrical Actuation Systems: Electrical systems, Mechanical switches, solid-state switches, solenoids, DC & AC motors, Stepper motors and their merits and demerits.

Signal Conditioning: Introduction to signal conditioning. The operational amplifier, Protection, Filtering, Wheatstone bridge, Digital signals Multiplexers, Data acquisition, Introduction to Digital system. Processing Pulse-modulation.

UNIT-III

Introduction to Microprocessors: Evolution of Microprocessor, Organization of Microprocessors (Preliminary concepts), basic concepts of programming of microprocessors.

Review of concepts - Boolean algebra, Logic Gates and Gate Networks, Binary & Decimal number systems, memory representation of positive and negative integers, maximum and minimum integers. Conversion of real, numbers, floating point notation, representation of floating point numbers, accuracy and range in floating point representation, overflow and underflow, addition of floating point numbers, character representation.

UNIT -IV

Logic Function: Data word representation. Basic elements of control systems 8085A processor architecture terminology such as CPU, memory and address, ALU, assembler data registers, Fetch cycle, write cycle, state, bus, interrupts. Micro Controllers. Difference between microprocessor and micro controllers. Requirements for control and their implementation in microcontrollers. Classification of micro controllers.

UNIT V

Organization & Programming of Microprocessors: Introduction to organization of INTEL 8085-Data and Address buses, Instruction set of 8085, programming the 8085, assembly language programming.

Central Processing Unit of Microprocessors: Introduction, timing and control unit basic concepts, Instruction and data flow, system timing, examples of INTEL 8085 and INTEL 4004 register organization.

REFERENCE BOOKS:

1. Mechatronics, W.Bolton, Longman, 2Ed, Pearson Publications,2007.
2. Microprocessor Architecture, Programming And ApplicationsWith 8085/8085A, R.S. Ganokar, Wiley Eastern.
3. Mechatronics and Microprocessors, K.P.Ramchandran, G.K.Vijayraghavan, M.S.Balasundran, Wiley, 1st Ed, 2009
4. Mechatronics - Principles, Concepts and applications – Nitaigour and Premchand Mahilik - Tata McGraw Hill- 2003.
5. Mechatronics Principles & applications, Godfrey C. Onwubolu,Elsevier. .
6. Introduction Mechatronics & Measurement systems, David.G.
7. Aliciatore & Michael. B. Bihistaned, Tata McGraw Hill, 2000.

MAINTENANCE ENGINEERING & MANAGEMENT

Unit-I

Introduction, operating life cycle, reliability, Failure data analysis, failure rate curve, hazard models, elements in series, parallel, mix, logic diagrams, improving reliability, redundancy-element, unit, standby, maintainability, availability, reliability and maintainability trade off.

Unit-II

Maintenance Strategies: Break down maintenance, planned maintenance, strategies, preventive maintenance, design out maintenance, planned lubrication, total productive maintenance, zero break down, preventive inspection of equipment used in emergency.

Unit-III

Replacement planning maintain or replace decision, replacement of items that deteriorate identical equipment, replacement of items that fail without deterioration individual, group replacement, replacement in anticipation of failure.

Unit-IV

Break down maintenance planning, assignment model, waiting time models expected waiting time, minimum cost service rate, PERT.

Unit-V

Maintenance Management, production maintenance system, objectives and functions, forms, policy, planning, organization, economics of maintenance, manpower planning, materials planning, spare parts planning and control, evaluation of maintenance management.

Books:

1. Management of systems – R.N. Nauhria & R. Prakash.
2. Operations Research – Wangner.
3. Engineering Management, S.C. Sharma & T.R. Banga, Khanna Book Publishing Co.

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AUTOMOBILE ENGINEERING

EVALUATION SCHEME (CBCS)

VI SEMESTER

Sl No.	Subject Code	Subject Name	Teaching Deptt.	L-T-P	Th/ Lab Marks	Sessional		Total	Credit
						Test	Assig/Att		
1	RAS 601	INDUSTRIAL MANAGEMENT	Applied Science	3---0---0	70	20	10	100	3
2	RUC 601/ RAS 602	CYBER SECURITY/ SOCIOLOGY	Applied Science	3---0---0	70	20	10	100	3
3	RAU 601	Automotive Fuels and Lubricants	ME.	3---0---0	70	20	10	100	3
4	RME 602	Theory of Machines	ME.	3---1---0	70	20	10	100	4
5	RAU 602	Automotive chassis and suspension	ME ME	3---0---0	70	20	10	100	3
6	RAU 061,062,063, RME063	**Department Elective Course –2**	ME	3---1---0	70	20	10	100	4
7	RAU651	Automotive Fuels and Lubricants Lab	ME	0---0---2	50		50	100	1
8	RME652	Theory of Machines Lab	ME	0---0---2	50		50	100	1
9	RAU652	Design & Simulation Lab-II	ME	0---0---2	50		50	100	1
10	RME654	Refrigeration & Air-conditioning Lab	ME	0--0--2	50		50	100	1

Department Elective Course –2

RAU061 : Vehicle Dynamics
RAU062 : Tribology
RAU063 : Manufacturing of Automotive Components
RME063 : Mechanical vibrations

AUTOMOTIVE FUELS AND LUBRICANTS

UNIT I

MANUFACTURE OF FUELS AND LUBRICANTS

Structure of petroleum refining process, classification of petroleum fuels, thermal cracking,catalytic cracking, polymerization, alkylationisomerisation, blending, products of refining process. Manufacture of lubricating oil base stocks, manufacture of finished automotive lubricants.

UNIT II

PROPERTIES & TESTING OF FUELS

Thermo-chemistry of fuels, properties and testing of fuels & Lubricants, relative density,calorific value, fire point, distillation, vapor pressure, flash point, spontaneous ignition,temperature, viscosity, pour point, flammability, ignitability, diesel index. API gravity, aniline point, Viscosity index etc.

UNIT III

FUEL RATING & ADDITIVES:

Fuels for SI and CI engine, important qualities of SI and CI engine fuels, Rating of SI engine and CI engine fuels, Dopes, Gaseous fuels

Additive - mechanism, requirements of an additive, petrol fuel additives and diesel fuel additives specifications of fuels, diesel knock, Cetane rating.

UNIT IV

ALTERNATE FUELS

Use of alternate fuel in engines- LPG. CNG need for alternate fuels, availability & theirproperties, general use of alcohols. LPG.CNG.LNG, hydrogen, ammonia, vegetable oils, biodiesel & biogas. merits & demerits of alternate fuels. Introduction to alternate energy sources like, electric vehicle, hybrid, fuel cell & solar cars.

UNIT V

LUBRICANTS

Classification of lubricating oils, properties of lubricating oils, tests on lubricants. Grease classification, properties, tests. Specific requirements for automotive lubricants, oxidation, deterioration and degradation of lubricants, additives, synthetic lubricants.

THEORY OF LUBRICANTS: Engine friction - introduction, total engine friction, effect of engine variables on friction, hydrodynamic lubrication, elasto hydrodynamic lubrication, boundary lubrication, bearing lubrication, functions of the lubrication system, introduction to design of a lubricating system.

TEXT BOOKS

1. Internal Combustion Engineering by Ganesan V. TaU McGraw -Hill Publishing Co. New Delhi.
2. Lubrication. Raymond G. Gunther, Chipton Book Co., 1971.

REFERENCEBOOKS

1. Fuels - Solids. Liquids.Gaseous by Brame.J.S.S. and King.I .G.
2. Fuels and Fuel Technology by Francis, W, Vol. I & II
3. Modern Petroleum Technology by Hobson, G.D. & Pohl. W
4. Elements of Fuels and Combustion, O.P. Gupta, Khanna Book Publishing Co.
5. Lubrication-A practical guide to lubricant selection by A.R. Lansdown. Pergamon press,1982.
6. Energy today & tomorrow by MaheswarDayal, I & B Horishr India.
7. Internal Combustion Engineering and Air Pollution by Oibert. E.F.: International Book Co 1988.
8. Automotive Engines, A.K. Babu, Khanna Publishing House

THEORY OF MACHINES

Unit – I

INTRODUCTION: DEFINITIONS: Link or element, kinematic pairs, degrees of freedom, Grubler's criterion (without derivation), Kinematic chain, Mechanism, structure, Mobility of Mechanism, Inversion, Machine Gashoff's criteria.

Kinematic Chains And Inversions: Inversions of Four bar chain; Single slider crank chain and Double slider crank chain.

MECHANISMS: Quick return motion mechanisms-Drag link mechanism, Whitworth mechanism and Crank and slotted lever Mechanism. Straight line motion mechanisms Peaucellier's mechanism and Robert's mechanism. Intermittent Motion mechanisms Geneva mechanism and Ratchet and Pawl mechanism. Toggle mechanism, Pantograph, Davis & Ackerman steering gear mechanism.

Unit II

CAMS: Types of cams, Types of followers, Displacement, Velocity and Acceleration time curves for cam profiles. Disc cam with reciprocating follower having knife-edge, roller and flat-faced follower, Disc cam with oscillating roller follower, Follower motions including SHM, Uniform velocity, uniform acceleration and retardation and Cycloidal motion.

Unit III

BALANCING OF MACHINERY: Static and Dynamic balancing, balancing of singlerotating mass in same plane and in different planes. Balancing of several rotating masses in same plane and in different planes. Balancing of reciprocating masses. Inertia effect of
crank and connecting road.

Unit - IV

SPUR GEARS: Gear terminology, law of gearing, Characteristics of involute action, Path of contact, Arc of contact, Contact ratio, Interference in involute gears, Methods of avoiding: interference, Back lash, Comparison of involute and cycloidal teet

GEAR TRAINS: Simple gear trains, Compound gear trains for large speed reduction, Epicyclic gear trains, Algebraic and tabular methods of finding velocity ratio of epicyclic gear trains. Tooth load and torque calculations in epicyclic gear trains.

Unit V

BELTS, ROPES AND CHAINS: Introduction, Belt and Rope drives, open and crossed belt drives, action of belt on pulleyes, velocity ratio, slip, law of belting, length of belt, Ratio of friction, chains, chain length and analytical, classification of chains.

GOVERNORS: Types of governors; force analysis of Porter and Hartnell governors. Controlling force, stability, sensitiveness, isochronisms, effort and power.

Text Books:

1. Rattan S.S, “Theory of Machines” Tata McGraw-Hill Publishing Company Ltd., New Delhi, and 2nd edition -2005.
2. Sadhu Singh, “Theory of Machines,” Pearson Education (Singapore) Pvt. Ltd., Indian Branch, New Delhi, 2ND Edi. 2006.
3. Jagadish Lal, 'Theory of Machine', Dhanpat Rai Publications, New Delhi.

Reference books:

1. Shigley. J. V. and Uickers, J.J., “Theory of Machines & Mechanisms” OXFORD University press.2004
2. “Theory of Machines -I”, by A.S.Ravindra, Sudha Publications, Revised 5th Edi. 2004.

AUTOMOTIVE CHASSIS AND SUSPENSION

Unit I

Automotive chassis: Definition; chassis layout; types of chassis layout with reference to power plant location, steering position and drive on wheels; chassis components; chassis classification; Automotive frames: Construction; functions; loads acting; materials; types; frame cross sections; frame diagnosis and service; dimensions of wheel base; wheel track; chassis overhang and ground clearance.

Unit II

Front axle & steering system: Functions, construction & types of front axle; front wheel geometry; front wheel drive; steering mechanisms; steering linkages & layout; types of steering gear boxes; power & power assisted steering; electronic steering; four-wheel steering; terminology-reversible steering, under-steering, over-steering, turning radius.

Unit III

Suspension system: Need; factors influencing ride comfort; types; suspension springs-leaf spring, coil spring & torsion bar; spring materials; independent suspension; rubber suspension; pneumatic suspension; hydraulic suspension, shock absorbers-liquid & gas filled.

Unit IV

Braking systems: Introduction, principles of braking; classification; brake actuating mechanisms; Drum brake- theory; principle; construction; working; Disc brake- theory, principle, construction, working; Parking brake- theory, principle; construction, types; Hydraulic system theory, principles, master-cylinder basics, wheel-cylinder basics, tubing & hoses, valves & switches, brake fluid; Powerbrake- theory, vacuum-booster basics, hydraulic-booster basics, electro-hydraulic booster basics; Advanced brake theories; Exhaust brake; abs technology; factors affecting brake performance operating temperature, area of brake lining, clearance.

Unit V

Wheel: Forces acting on wheels, construction of wheel assembly, types- spoke, disc & built-up wheels; wheel balancing; wheel alignment; Tyres: Static & rolling properties of tyres, construction details, types of tyres- pneumatic & hydraulic; types of tyre-wear & their causes; tyre rotation.

Bearings: Functions; classification of bearings; bearing materials; automotive bearings.

Books and References:

1. Automobile engineering", Dr. Kripal Singh.
2. A.K. Babu, Automotive Mechanics, Khanna Publishing House
3. Automobile engineering" K.M. Gupta.
4. Heldt P.M., "Automotive chassis", Chilton Co., New York.
5. Giles J.G., "Steering, Suspension and tyres", Iliffe Book Co., London.

AUTOMOTIVE FUELS AND LUBRICANTS LAB

Minimum eight experiments out of the followings:

1. Temperature dependence of viscosity of lubrication oil by Redwood Viscometer.
2. Viscosity Index of lubricating oil by Saybott Viscometer.
3. Flash and Fire points of Diesel, K-Oil, Bio Diesel.
4. Flash and Fire points of lubricants.
5. Drop point of grease and mechanical penetration in grease.
6. Calorific value of liquid fuel.
7. Calorific value of gaseous fuel
8. Study of semi solid lubrication in various Automobile Unit & Joints
9. Study of lubrication in transmission, final drive, steering gearbox.
10. Study of analytical equipment for oil analysis.
11. To find out volatility characteristic of different fuels by ASTM distillation methods (diesel, gasoline lubricants).

THEORY OF MACHINES LAB

Note: Minimum Ten experiments out of the following:

1. Study of simple linkage models/mechanisms
2. Study of inversions of four bar linkage
3. Study of inversions of single/double slider crank mechanisms
4. Experiment on Gears tooth profile, interference etc.
5. Experiment on Gear trains
6. Experiment on longitudinal vibration
7. Experiment on transverse vibration
8. Experiments on dead weight type governor
9. Experiment on spring controlled governor
10. Experiment on critical speed of shaft
11. Experiment on gyroscope
12. Experiment on static/dynamic balancing
13. Experiment on Brake
14. Experiment on clutch

DESIGN AND SIMULATION LAB-II

- Computer and Language: Students are required to refresh the basics of computer language such as C++ or MATLAB so that they should be able to write the computer program.
- Writing Computer programme for conventional design: Students are required to write computer program and validate it for the design of machine components done in theory subject .
- Mini Project: Based from real life applications and optimize using modelling and analysis software
- Students will take up a real life problem for design of a subsystem/system using modelling & Analysis software tools. This will be done as assignment in groups to be submitted at the end of the semester.

REFRIGERATION & AIR CONDITIONING Lab **L T P**
0 0 2

Minimum eight experiments out of the following:

1. Experiment on refrigeration test rig and calculation of various performance parameters.
2. Study of different types of expansion devices used in refrigeration system.
3. Study of different types of evaporators used in refrigeration systems.
4. To study basic components of air-conditioning system.
5. Experiment on air-conditioning test rig & calculation of various performance parameters.
6. Experiment on air washers
7. Study of window air conditioner.
8. Study & determination of volumetric efficiency of compressor.
9. Visit of a central air conditioning plant and its detailed study.
10. Visit of cold-storage and its detailed study.
11. Experiment on Ice-plant.
12. Experiment on two stage Reciprocating compressor for determination of volumetric efficiency , PV diagram and effect of intercooling.
13. Study of Hermetically sealed compressor.
14. Experiment on Desert coolers.

Department Elective Course –2

1. Vehicle Dynamics
2. Tribology
3. Manufacturing of Automotive Components
4. Mechanical vibrations
5. Advanced welding Technology

VEHICLE DYNAMICS

UNIT I

CONCEPT OF VIBRATION

Definitions, Modeling and Simulation, Global and Vehicle Coordinate System, Free, Forced, Undamped and Damped Vibration, Response Analysis of Single DOF, Two DOF, Multi DOF, Magnification factor, Transmissibility, Vibration absorber, Vibration measuring instruments, Torsional vibration, Critical speed.

UNIT II

TIRES

Tire forces and moments, Tire structure, Longitudinal and Lateral force at various slip angles, rolling resistance, Tractive and cornering property of tire. Performance of tire on wet surface. Ride property of tires. Magic formulae tire model, Estimation of tire road friction. Test on Various road surfaces. Tire vibration.

UNIT III

VERTICAL DYNAMICS

Human response to vibration, Sources of Vibration. Design and analysis of Passive, Semi-active and Active suspension using Quarter car, half car and full car model. Influence of suspension stiffness, suspension damping, and tire stiffness. Control law for LQR, H-Infinite, Skyhook damping. Air suspension system and their properties.

UNIT IV

LONGITUDINAL DYNAMICS AND CONTROL

Aerodynamic forces and moments. Equation of motion. Tire forces, rolling resistance, Load distribution for three wheeler and four wheeler. Calculation of Maximum acceleration, Reaction forces for Different drives. Braking and Driving torque. Prediction of Vehicle performance. ABS, stability control, Traction control.

UNIT V

LATERAL DYNAMICS 9

Steady state handling characteristics. Steady state response to steering input. Testing of handling characteristics. Transient response characteristics, Direction control of vehicles. Roll center, Roll axis, Vehicle under side forces. Stability of vehicle on banked road, during turn. Effect of suspension on cornering.

TEXT BOOKS:

1. Singiresu S. Rao, "Mechanical Vibrations", 5th Edition, Prentice Hall, 2010
2. Wong. J. Y., "Theory of Ground Vehicles", 3rd Edition, Wiley-Interscience, 2001

3. Rajesh Rajamani, "Vehicle Dynamics and Control", 1st edition, Springer, 2005
4. Thomas D. Gillespie, "Fundamentals of Vehicle Dynamics", Society of Automotive Engineers Inc, 1992

REFERENCES:

1. Dean Karnopp, "Vehicle Stability", 1st edition, Marcel Dekker, 2004
2. Nakhaie Jazar. G., "Vehicle Dynamics: Theory and Application", 1st edition, Springer, 2008
3. Michael Blundell & Damian Harty, "The Multibody Systems Approach to Vehicle Dynamics", Elsevier Limited, 2004
4. Hans B Pacejka, "Tire and Vehicle Dynamics", 2nd edition, SAE International, 2005
5. A.K. Babu, Hybrid Electric Vehicles, Khanna Publishing House
6. John C. Dixon," Tires, Suspension, and Handling", 2nd edition, Society of Automotive Engineers Inc, 1996

TRIBOLOGY

UNIT-I

Study of various parameters: Viscosity, flow of fluids, viscosity and its variation-absolute and kinematic viscosity, temperature variation, viscosity Index determination of viscosity, different viscometers used.

Hydrostatic lubrication: Hydrostatic step bearings, applications to pivoted pad thrust bearing and other applications, hydrostatic lifts, hydrostatic squeeze films and its application to journal bearings.

UNIT II

Hydrodynamic theory of lubrication: Various theories of lubrication, petroffs equation, Reynold's equation in two dimensions -Effects of side leakage - Reynolds equation in three dimensions, Friction in sliding bearing, hydro dynamic theory applied to journal bearing, minimum oil film thickness, oil whip and whirl anti –friction bearing.

UNIT-III

Friction and power losses in journal bearings: Calibration of friction loss friction in concentric bearings, bearing modulus, Somraerfield number, heat balance, practical consideration of journal bearing design considerations.

UNIT-IV

Air lubricated bearing: Advantages and disadvantages application to Hydrodynamic journal bearings, hydrodynamic thrust bearings. Hydrostatic thrust bearings. Hydrostatic bearing Analysis including compressibility effect. Study of current concepts ofboundary friction and dry friction.

UNIT-V

Types of bearing oil pads: Hydrostatic bearing wick oiled bearings, oil rings, pressure feed bearing, partial bearings -externally pressurized bearings.

Bearing materials: General requirements of bearing materials, types of bearing materials.

TEXT BOOKS:

1. Fundamentals of Tribotogy, Basu, SenGupta and Ahuja, PHI

2. Tribotogy in Industry: Sushil Kumar Srivatsava, S. Chand &Co. REFERENCE:

1.Tribology - B.C. Majumdar

MANUFACTURING OF AUTOMOTIVE COMPONENTS

UNIT I

CASTED ENGINE COMPONENTS

Material selection and Manufacturing methods for Piston, Piston rings, Cylinder block, wet and dry liners, Engine head, Oil pan, Carburetors. Thermal barrier coating of Engine head and valves.

UNIT II

FORGED ENGINE COMPONENTS

Material selection and Manufacturing methods for Crank shaft, Connecting rod, Cam shaft, valve, Piston pin, Push rod, Rocker arm, tappets, spark plug.

UNIT III

TRANSMISSION SYSTEM

Material selection and Manufacturing methods for Clutch – Clutch lining – Gear Box – Gear – Propeller Shaft – Differential – Axle Shaft – Bearing – fasteners – Wheel drum. Methods of Gear manufacture – Gear hobbing and gear shaping machines - gear generation – gear finishing and shaving – Grinding and lapping of hobs and shaping cutters – gear honing – gear broaching.

UNIT IV

VEHICLE CHASSIS

Material selection and manufacturing methods for chassis, dead axle, leaf spring, coil spring and shock absorbers – wheel housing – steering system, Brake shoes, wheel rim, Tyres. Heat treatment procedures.

UNIT V

RECENT DEVELOPMENTS

Surface treatment – Plastics – Plastics in Automobile vehicles – Processing of plastics - Emission control system – catalytic converter – Hydro forming of exhaust manifold and lamp housing – stretch forming of Auto body panels – MMC liners –Selection of materials for Auto components. Use of Robots in Body weldment.

TEXT BOOK:

1. Heldt.P.M, "High speed combustion engines", Oxford publishing Co., New York, 1990.

REFERENCES:

1. Kirpal Singh, 'Automobile Engineering", Vol. I & II, Standard Publishers, New Delhi, 1997.
2. A.K. Babu, Automotive Mechanics, Khanna Publishing House
3. Newton and steels, the motor vehicle, ELBS, 1990
4. Serope Kalpakjian and Steven R. Schmid, "Manufacturing Processes for Engineering
5. Materials", Fourth Edition, Pearson Education publications – 2003
6. Gupta K.M. "Automobile Engineering" Vol.I & II, Umesh Publishers, 2000.

MECHANICAL VIBRATION

UNIT – I

Introduction: Periodic motion, harmonic motion, superposition of simple harmonic motions, beats, Fourier analysis.

Single Degree Freedom System

Free vibration, Natural frequency, Equivalent systems, Energy method for determining natural frequency, response to an initial disturbance, Torsional vibrations, Damped vibrations, Vibrations of systems with viscous damping, Logarithmic decrement

UNIT – II

Single Degree Freedom: Forced Vibration

Forced vibration, Harmonic excitation with viscous damping, steady state vibrations, Forced vibrations with rotating and reciprocating unbalance, Support excitation, Vibration isolation, Transmissibility, Vibration measuring instruments, Displacement, velocity and acceleration measuring instruments

UNIT- III

Two Degree Freedom systems

Introduction, Principal modes, Double pendulum, Torsional system with damping, coupled system, undamped dynamic vibration absorbers, Centrifugal pendulum absorbers, Dry friction damper

UNIT- IV

Multi Degree Freedom system: Exact Analysis

Undamped free and forced vibrations of multi-degree freedom systems, influence number, Reciprocal theorem, Torsional vibration of multi-degree rotor system, Vibration of gear system, Principal coordinates, Continuous systems- Longitudinal vibrations of bars, Torsional vibrations of circular shafts

UNIT- V

Multi Degree Freedom system: Numerical Analysis

Rayleigh's, Dunkerely's, Holzer's ad Stodola methods, Rayleigh-Ritz
method Critical speed of shafts
Shaft with one disc with and without damping, Multi-disc shafts, Secondary critical speed.

Books and References:

1. Mechanical Vibrations – P. Srinivasan, TMH
2. Mechanical Vibrations – G. K. Groover, Jain Brothers, Roorkee
3. Mechanical Vibrations – JS Rao & K Gupta, New Age
4. Mechanical Vibrations-Theory & Practice, S Bhave, Pearson Education.
5. Mechanical Vibrations-Theory & Applications, Singhal, Katson Books.
6. Theory of Vibrations with Applications, Thomson&Dahleh, Pearson Education.
7. Elements of Vibration Analysis, L Meirovitch, McGraw-Hill Education.
8. Mechanical Vibrations – Tse, Morse & Hinkle
9. Mechanical Vibrations – V. Rama Murthy, Narosa Publications
10. Mechanical Vibrations – D. Nag, Wiley

ADVANCED WELDING TECHNOLOGY

Unit-I

Introduction : Importance and application of welding, classification of welding process. Selection of welding process.

Brief review of conventional welding process : Gas welding, Arc welding, MIG, TIG welding. Resistance welding. Electroslag welding, Friction welding etc. Welding of MS.CI, Al, Stainless steel & Maurer/Schaefflar Diagram. Soldering & Brazing.

Unit-II

Advanced welding Techniques- Principle and working and application of advanced welding techniques such as Plasma Arc welding, Laser beam welding, Electron beam welding, Ultrasonic welding etc.

Unit-III

Advanced welding Techniques (continued) : Principle and working and application of advanced welding techniques such as explosive welding/ cladding, Underwater welding, Spray-welding / Metallising, Hard facing.

Unit-IV

Weld Design : Welding machines/equipments and its characteristics and arc-stability, Weld defects and distortion and its remedies, Inspection/testing of welds, Weld Design, Welding of pipe-lines and pressure vessels. Life predication.

Unit-V

Thermal and Metallurgical consideration.: Thermal considerations for welding, temperature distribution, Analytical/Empirical analysis/formulae, heating & cooling curves. Metallurgical consideration of weld, HAZ and Parent metal, micro & macro structure. Solidification of weld and properties.

Books

1. Welding Hand Book
2. Metal Casting and Joining - John.K.C- PHI Publications
3. Welding & Welding Technology- Richard L Little. Mc Graw Hill
4. Welding Principles and Practices – EdwardR. Bohnart , Mc Graw Hill, 4th Edition.